Used-car price estimation in Kazakhstan



Machine Learning Algorithms
Endterm course project

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Problem, data and cleaning

Advertised asking price

Supervised regression in KZT
Kolesa-Cars-2025, Kaggle
Vehicle specifications available in each ad

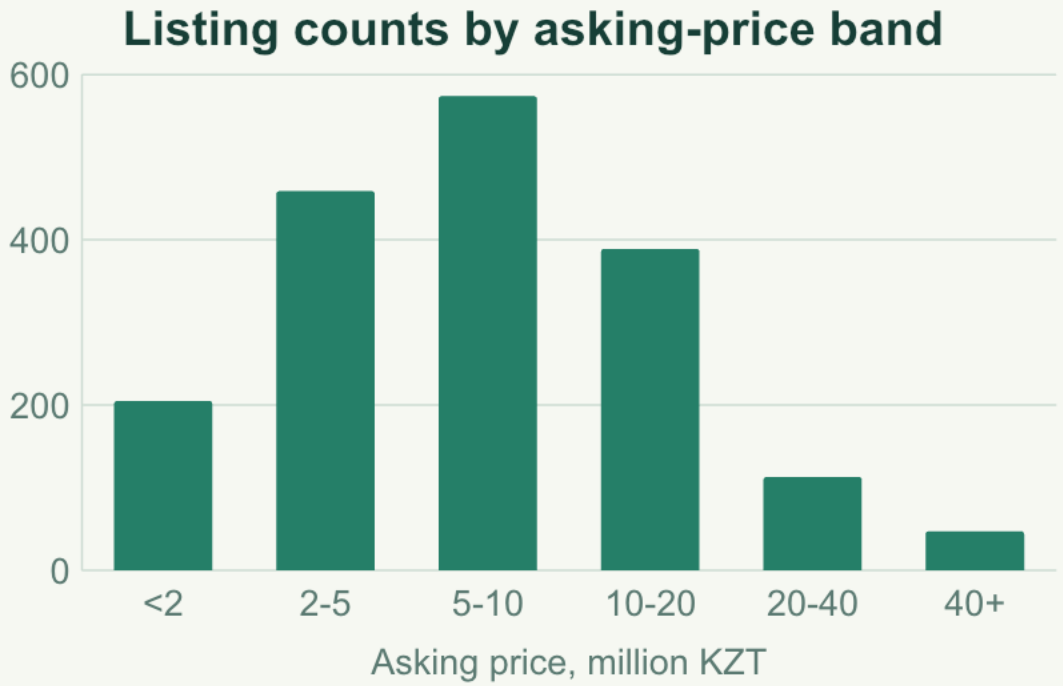
2,553

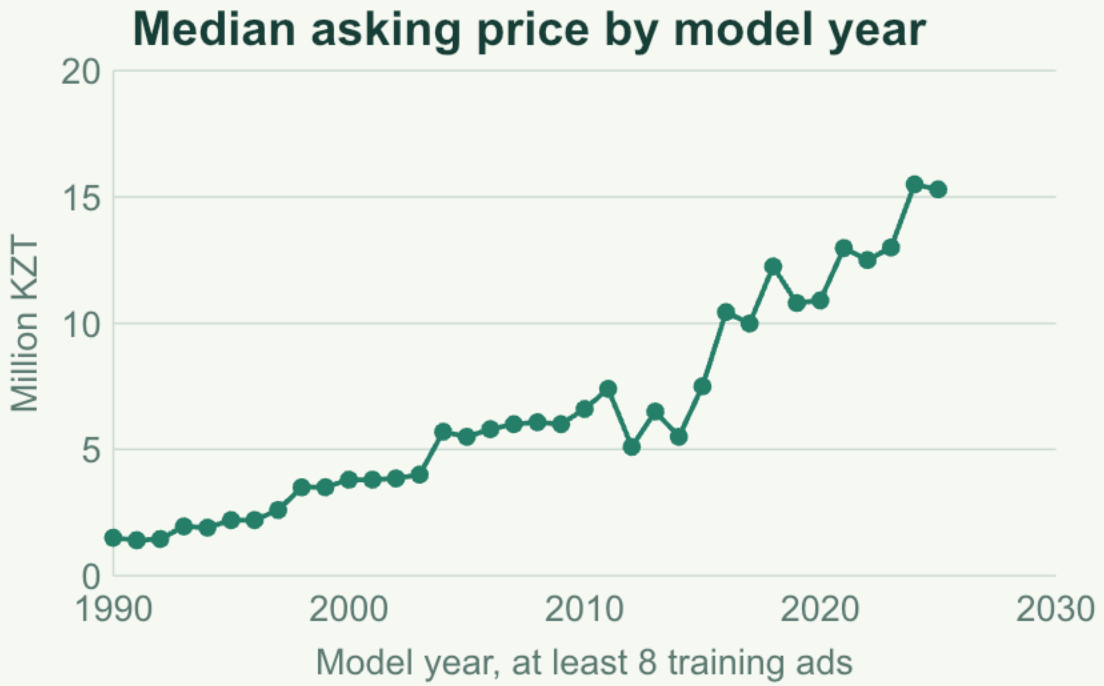
unique used-car listings
38-column derived export, original cells retained

Cleaning step	Rows
Original source	3,999
New cars removed	527
On-order ads removed	51
Repeated used ads removed	868
Retained	**2,553**

Goal: at least 20% lower test MAE than the training-median baseline

Training EDA: price and model year





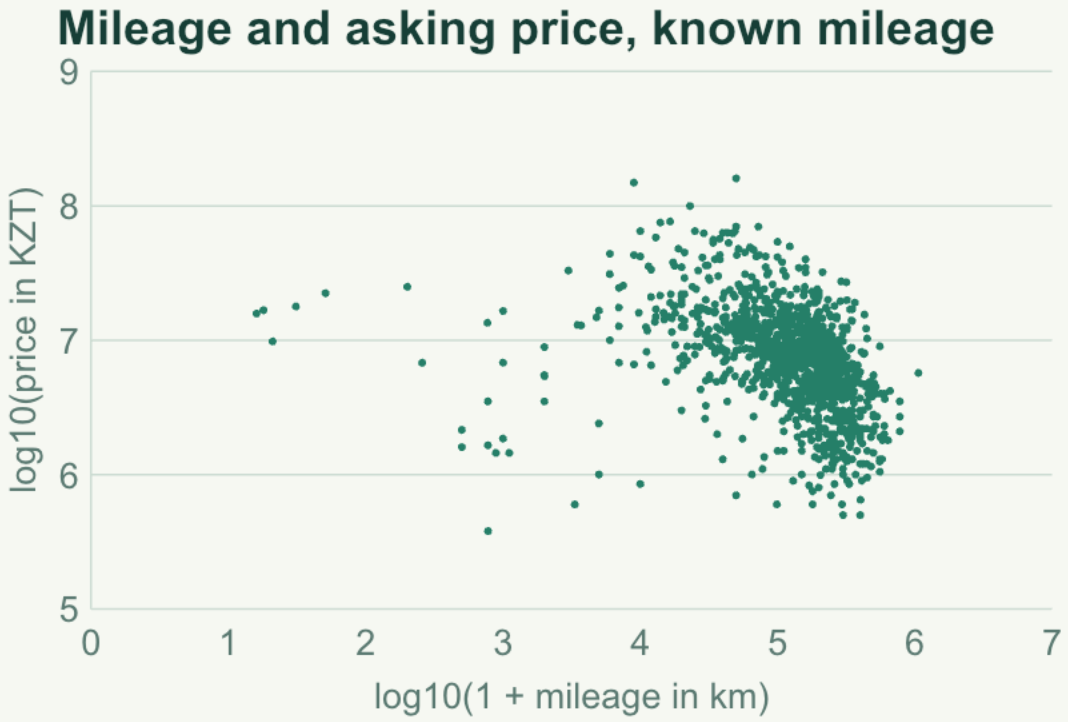
Mean 9.53m > median 6.70m KZT

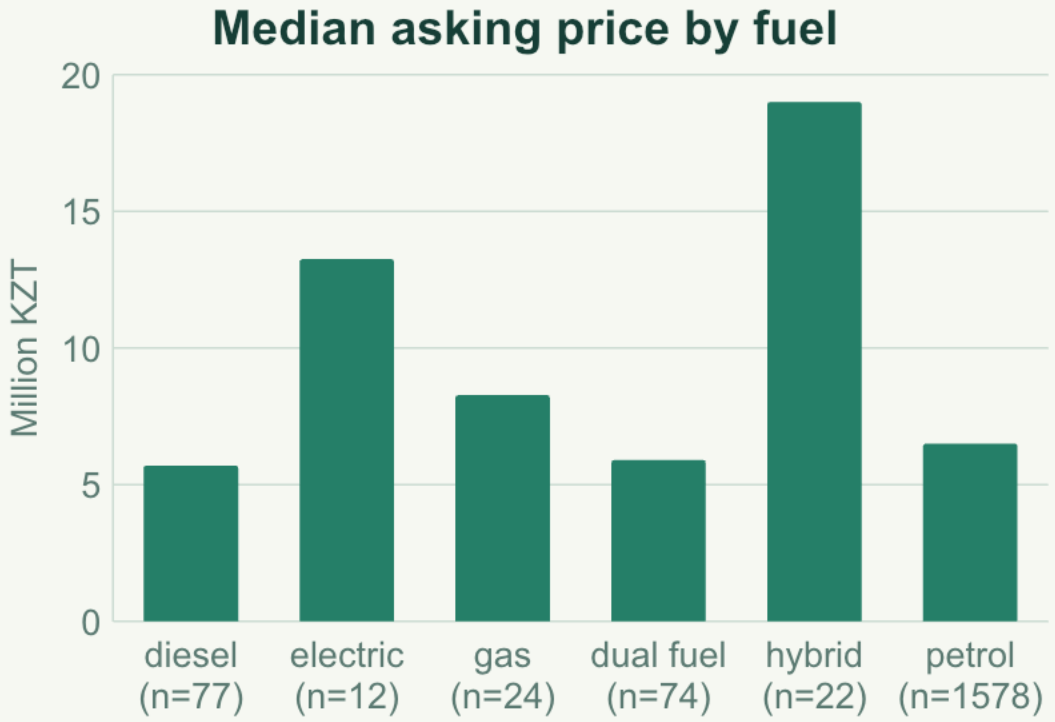
A small high-price tail raises the mean

Year-price Spearman ρ = 0.69

Newer years generally have higher median prices

Training EDA: mileage and fuel





Mileage-price ρ = -0.53

Higher mileage accompanies lower prices
Seeded sample, n = 1200

Fuel groups also differ in vehicle mix

Electric and hybrid samples are small
Group differences are descriptive

Preprocessing and evaluation protocol

1,787
Training, 70%

383
Validation, 15%

383
Historical test, 15%

Vehicle inputs

Year, log mileage, engine
Missing and EV applicability flags
Brand, model, body, fuel, gearbox

Feature processing

Train medians and standardization
Unknown category and OHE
Steering, fixed-2025 age proxies

Model decisions

3-fold train CV for tuned models
Validation ANN checkpoints
Fixed final ensemble recipe

Deduplicate before splitting. Fit learned preprocessing within each training fold.

Historical test feedback informed development. Results are retrospective.

Baselines and supervised model tuning

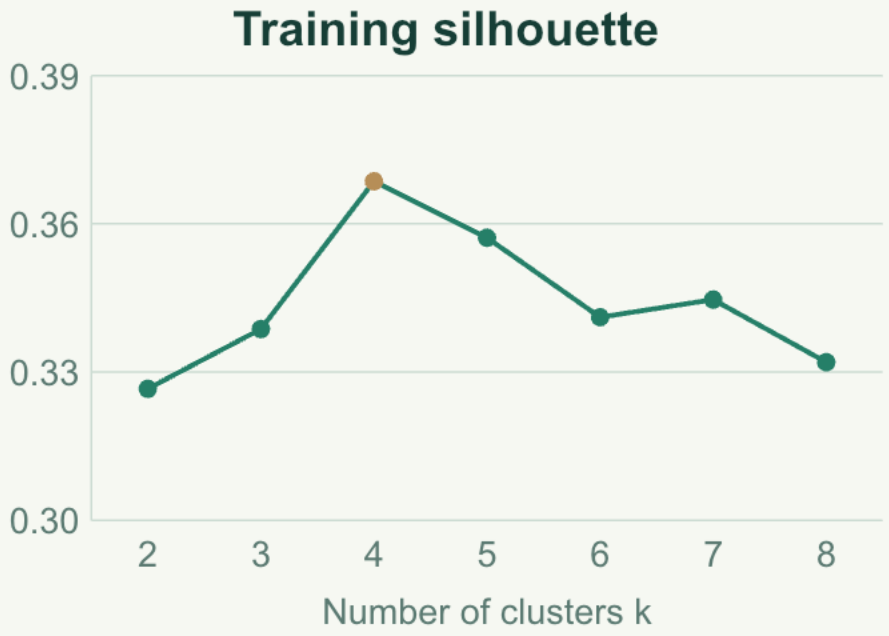
Original controls: median, Linear, KNN (k = 15, distance), Tree (depth 10, leaf 3)

Tuned model	Selected parameters	CV MAE ± SD	Val MAE
KNN tuned	k = 5, distance, p = 1 OHE min frequency = 1	2.62 ± 0.10	3.41
Tree tuned	depth = 10, min leaf = 3 OHE min frequency = 1	2.84 ± 0.14	3.36
RF tuned	**200 trees, max features = 0.7 min leaf = 1, OHE frequency = 1**	**2.17 ± 0.14**	**2.70**

Shuffled 3-fold KFold, seed 42. Minimize mean fold MAE in original KZT.

Values are million KZT. Fold SD describes variability, without a confidence claim.

K-Means selection and cluster profiles



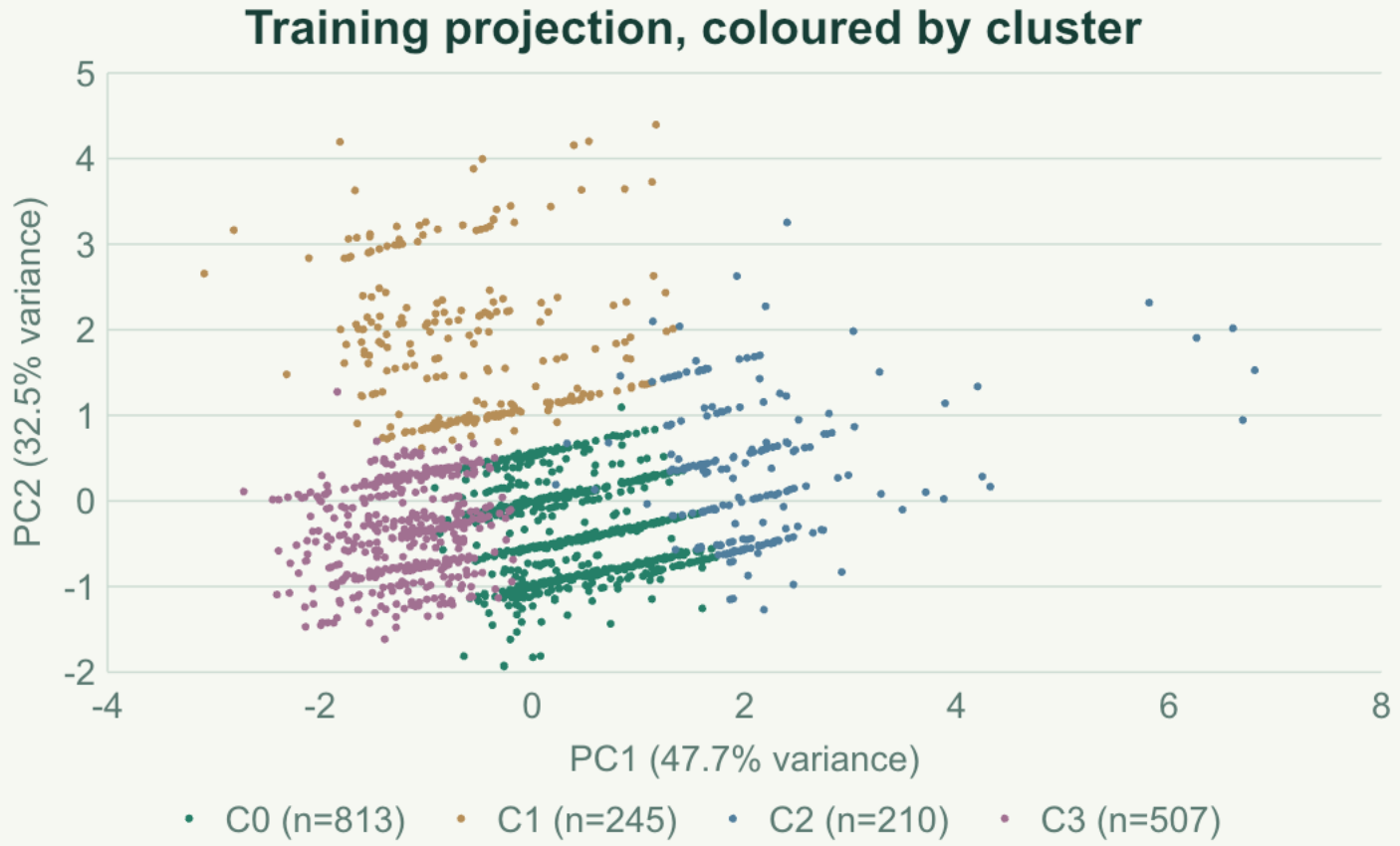
Cluster / n	Year	km, k	L	KZT, m
C0 / 813	2015	146.0	2.0	7.69
C1 / 245	2011	183.5	4.0	12.20
C2 / 210	2023	23.9	2.0	14.13
C3 / 507	1999	280.0	2.4	3.00

k = 4, silhouette = 0.3686

C3: oldest cars. C2: newest years. C1: largest median engine.

Medians shown. Fit: 1,775 non-electric ads, year/log km/engine, without price.

PCA of numeric vehicle features



80.13%

of numeric variance retained by two components

PC1 rises with year and falls with log mileage

PC2 mostly reflects engine size

19.9% remains outside this view. PCA is descriptive and is absent from the predictor.

Single ANN architecture and training

305
Encoded inputs

128
Hidden, ReLU

64
Hidden, ReLU

1
Linear output

Inputs and target

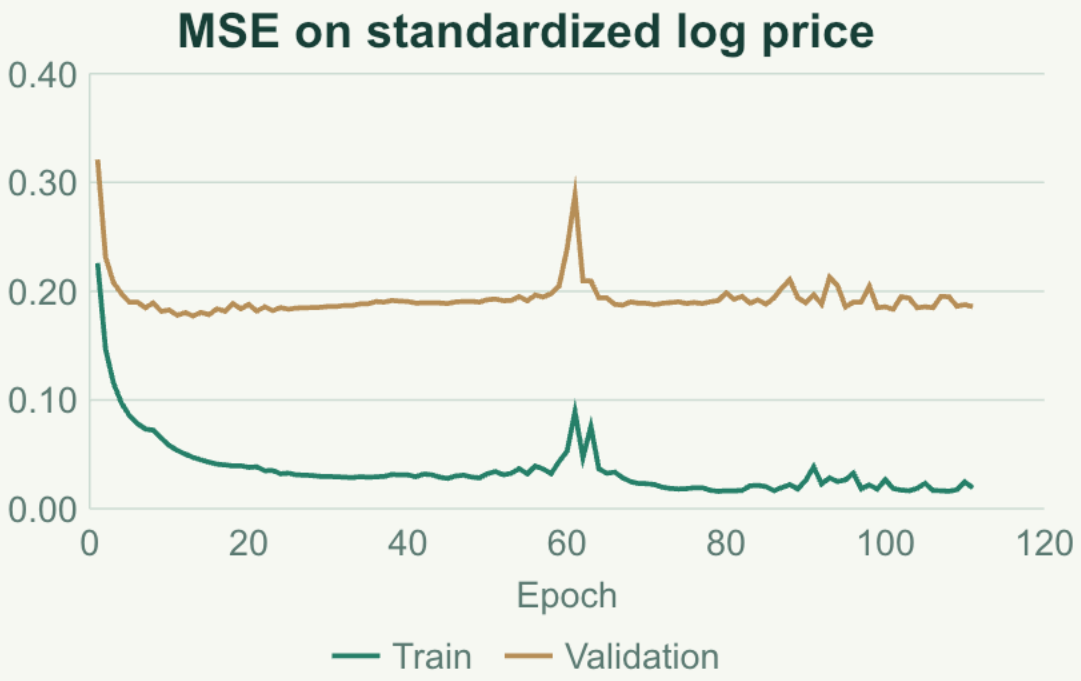
Corrected vehicle features, OHE frequency 2
Train-standardized log1p(price)
Inverse transform to non-negative KZT

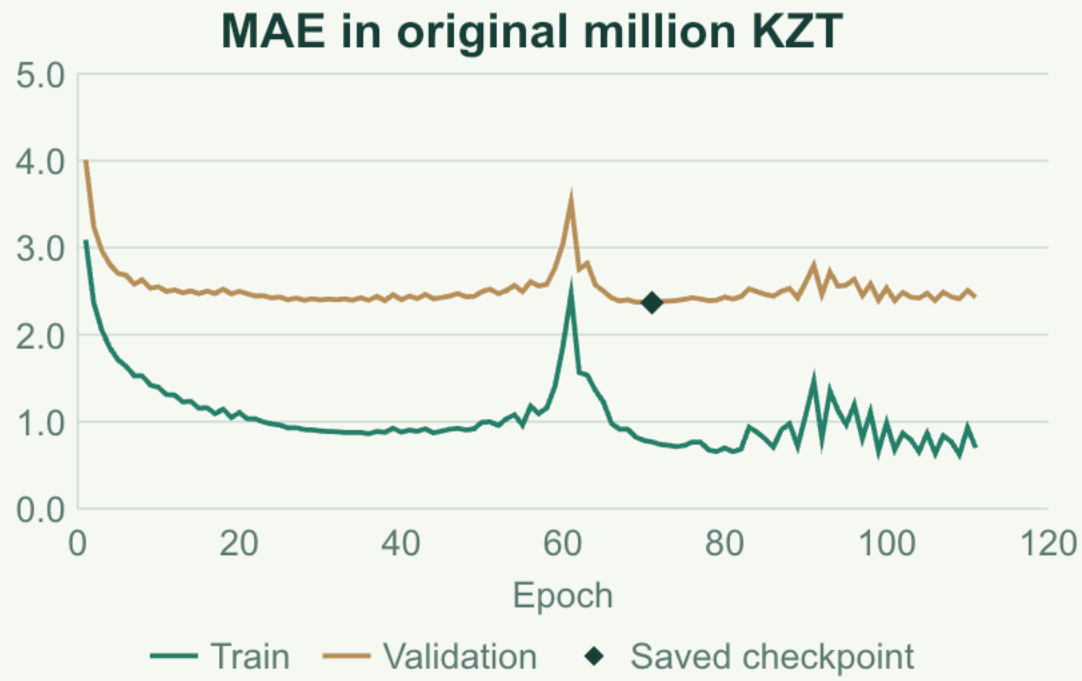
Backpropagation with Adam

Learning rate 0.001, batch 64, L2 α = 0.001
Seed 42, maximum 400 epochs
Validation MAE checkpoint, patience 40

Two fixed candidates: 64 / 32 and 128 / 64. Validation MAE selects architecture and epoch.

Single ANN learning curves and overfitting





Saved epoch 71 of 111. Validation MAE 2.37m KZT.

Final validation MAE is 2.5% above the saved best. A train-validation gap remains.

Final ANN ensemble: fixed prediction recipe

$\hat{y} = 0.25\,U + 0.5625\,C + 0.1875\,RF$

Component	Definition	Final weight
U: usage ANN pool	5 seeds, MAE checkpoints Steering + log age / annual km	25.00%
C: corrected ANN pool	5 seeds, dual-score checkpoints Corrected inputs + steering	56.25%
RF: tuned forest	200 trees, tuned vehicle features	18.75%

Every network uses 128 / 64 ReLU units. Seeds: 11, 23, 42, 71, 101.

Average inverse-transformed KZT predictions. Fixed recipe retained from prior development.

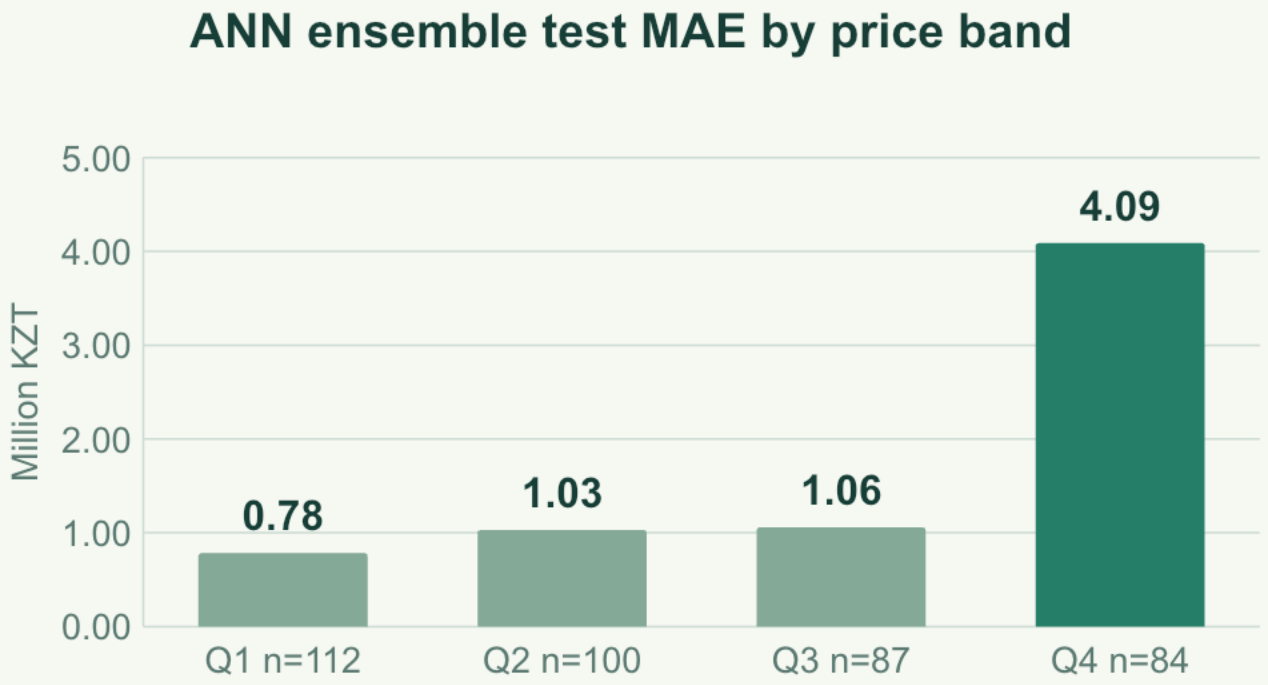
Final model comparison

Errors are million KZT. Same 383 validation and 383 historical-test listings.

Model	Val MAE	Val ±20%	Test MAE	Test ±20%	Test RMSE	Test R^2
Median	6.50	16.4%	5.95	17.8%	11.30	-0.048
Linear	4.65	32.9%	3.50	37.6%	6.46	0.657
KNN midterm	3.18	50.9%	2.44	54.3%	5.89	0.715
Tree midterm	3.44	46.7%	2.46	48.3%	5.48	0.753
KNN tuned	3.41	52.7%	2.79	47.8%	6.42	0.662
Tree tuned	3.36	45.7%	2.53	52.5%	6.21	0.684
RF tuned	2.70	55.1%	2.07	56.7%	4.97	0.797
ANN corrected	2.37	61.6%	1.74	50.9%	3.14	0.919
ANN ensemble	**2.24**	**67.6%**	**1.63**	**58.5%**	**3.19**	**0.917**

224 of 383 estimates (58.5%) fall within ±20% of the asking price.

Error analysis and limits



4.09m KZT

MAE in the highest price band

Sparse premium and unseen models
Asking prices, condition unobserved

Largest error: 2022 Mercedes-Benz EQS SUV

Actual 58.00m, predicted 30.21m KZT. Training quartiles define the bands.

Reused test cannot establish independent generalization. New later data is needed.

Conclusion and team responsibilities

72.5%

lower historical-test MAE
than the median baseline

MAE 1.63m KZT
RMSE 3.19m KZT
R^2 0.917, within ±20% 58.5%

Reproduce: python run_project.py
Saved source hash, split IDs, histories and predictor

Alisher Akhmet

Data sourcing, cleaning and feature engineering
EDA, K-Means and PCA

Aslan Muratov

Models, CV, ANN and ensemble
Metrics, errors and reproducibility

AI-assisted code and documentation
Each member reviews and defends their assigned material